WEI CHEN

CONTACT INFORMATION
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EDUCATION

University of Washington, Seattle, Washington
Ph.D, Molecular Engineering, 2015-2022
Field of Focus: Biological Engineering/Genome Science

Cornell University, Ithaca, New York
M.S., Applied and Engineering Physics, 2013-2015

ShanDong University, Jinan, ShanDong, China
B.S., Biological Science, 2008-2012

RESEARCH EXPERIENCE

University of Washington ..... *November 2022 to present*
Damon Runyon Postdoctoral Fellow *Advisor: David Baker*
- Designed synthetic transcription factor for gene regulation using generative models .

University of Washington ..... *March 2016 to October 2022*
Grad student *Advisor: Jay Shendure*
- Developed new genome engineering and molecular recording tools.
- Developed predictive models for CRISPR/Cas9-mediated double-strand break repair.

University of Washington ..... *September 2015 to March 2016*
Rotation student *Advisor: Georg Seelig*
- High-throughput single cell RNA profiling with combinatorial indexing.

Cornell University ..... *June 2014 to May 2015*
Grad Student *Advisor: Jan Lammerding*
- Used live cell imaging to study mechanotransduction through nuclear envelope.

Peking University ..... *September 2011 to July 2013*
Undergrad student *Advisor: Sunney.X. Xie and YuJie Sun*
- Built super-resolution imaging to study the role of nuclear actin in gene regulation.

PUBLICATIONS

co-first authors

1. Askary, A., **Chen, W.**, Choi, J., Du, L.Y., Elowitz, M.B., Gagnon, J.A., Schier, A.F., Seidel, S., Shendure, J., Stadler, T. and Tran, M., 2024. The lives of cells, recorded. **Nature Reviews Genetics**, (All authors contributed equally, with the order determined alphabetically by surename.)
2. **Chen, W.**, Choi, J., Nathans, J.F., Agarwal, V., Martin, B., Nichols, E., Leith, A., Lee, C. and Shendure, J., 2024. Symbolic recording of signalling and cis-regulatory element activity to DNA. **Nature**, 632, pp.1073-1081.
3. Li, X., **Chen, W.**, Martin, B.K., Calderon, D., Lee, C., Choi, J., Chardon, F.M., McDiarmid, T.A., Daza, R.M., Kim, H. and Lalanne, J.B., 2024. Chromatin context-dependent regulation and epigenetic manipulation of prime editing. **Cell**, 187(10), pp.2411-2427.
4. Choi, J., **Chen, W.**, Minkina, A., Chardon, F.M., Suiter, C.C., Regalado, S.G., Domcke, S., Hamazaki, N., Lee, C., Martin, B. and Daza, R.M., 2022. A time-resolved, multi-symbol molecular recorder via sequential genome editing. **Nature**, 608(7921), pp.98-107
5. Choi, J., **Chen, W.**, Suiter, C.C., Lee, C., Chardon, F.M., Yang, W., Leith, A., Daza, R.M., Martin, B. and Shendure, J., 2022. Precise genomic deletions using paired prime editing. **Nature Biotechnology**, 40(2), pp.218-226.
6. **Chen, W**., McKenna, A., Schreiber, J., Haeussler, M., Yin, Y., Agarwal, V., Noble, W.S. and Shendure, J., 2019. Massively parallel profiling and predictive modeling of the outcomes of

CRISPR/Cas9-mediated double-strand break repair. **Nucleic acids research**, 47(15), pp.7989-8003.
7. Klein, J.C., **Chen, W.**, Gasperini, M. and Shendure, J., 2018. Identifying novel enhancer elements with CRISPR-based screens. *ACS chemical biology*, 13(2), pp.326-332.
8. Rosenberg, A.B., Roco, C.M., Muscat, R.A., Kuchina, A., Sample, P., Yao, Z., Graybuck, L.T., Peeler, D.J., Mukherjee, S., **Chen, W**. and Pun, S.H., 2018. Single-cell profiling of the developing mouse brain and spinal cord with split-pool barcoding. **Science**, 360(6385), pp.176-182.

Talks & Posters

Genome Engineering: CRISPR Frontiers (*Talk*) ..... *August, 2022*
CSHL meeting, Cold Spring Harbor, NY, US
AGTD 2022 Annual meeting (*Talk*) ..... *July, 2022*
The Jackson Laboratory For Genomic Medicine, Farmington, CT, US
2022 ENCODE Consortium Meeting (*Talk, virtual*) ..... *March, 2022*
Stanford University, Stanford, CA, US
Genome Engineering Seminar Series (*Invited talk, virtual*) ..... *March, 2022*
Harvard Medical School, Cambridge, MA, US
9th annual Winter q-Bio Conference (*Talk*) ..... *February, 2022*
Oahu, HI, US
New York Genome Center (NYGC) (*Invited talk, virtual*) ..... *January, 2022*
New York, NY, US
Genome Engineering: CRISPR Frontiers (*Poster*) ..... *August, 2018*
CSHL meeting, Cold Spring Harbor, NY, US

Honors

Damon Runyon Postdoctoral Fellowship 300,000 $ ..... *2023 to 2027*

Teaching Experience

Teaching assistant, Cornell University
AEP 2640 Interfacing the Digital Domain with an Analog World ..... *Fall 2014*
PHYS 2213 Physics II: Electromagnetism ..... *Spring 2015*

Skills

Dry lab skills: Python, R. Mathematical modeling; Machine learning; NGS analysis; Single cell analysis; Webtool development.
Wet lab Skills: NGS library prep and sequencing; single cell RNA-seq; Cell culture and ES cell culture; Cloning; Single molecule techniques; Super-resolution fluorescence microscopy; Confocal microscopy.
Languages: Chinese (native), English (bilingual proficiency)